Data Cleaning and Analysis Report

User Behavior and Mobile Usage Analysis

1. Data Preparation and Cleaning

The first step of the project involved preparing the dataset to ensure its quality and reliability for analysis. The data processing was performed using Python and several libraries from the data science ecosystem, including Pandas, Matplotlib, Seaborn, and Scikit-learn.

To understand the structure of the dataset, the functions `df.info()` and `df.describe()` were used. These functions provided an overview of the variables, their data types, and the statistical distribution of numerical features. This step confirmed that the dataset was structurally consistent and contained no missing values that would require imputation.

During this phase, the dataset was also reviewed for possible inconsistencies and extreme values. Basic data validation ensured that the variables such as screen time, battery drain, and data usage were within reasonable ranges.

Exploratory Data Analysis (EDA) was then conducted to better understand the relationships between different variables. Several visualizations were created, including histograms, scatter plots, and joint plots. These visualizations helped identify patterns between screen time, battery consumption, and application usage.

A correlation heatmap was also generated to measure the strength of relationships between numerical variables. This allowed us to identify the most influential features affecting user behavior and device performance.

2. Feature Engineering

After preparing the dataset, several new variables were created to enhance the analysis and capture behavioral patterns more effectively.

A **Digital Burnout Score** was introduced to estimate the potential fatigue associated with prolonged device usage. This score combines information from screen time, application usage, and battery consumption to represent overall digital strain.

To improve data quality, statistical outlier detection was applied using the Z-score method. This helped identify users with extremely high or unusual usage behavior. These outliers were analyzed separately to ensure they did not distort the overall results.

Additionally, a **weighted risk score** was designed to categorize users based on their level of device engagement. The scoring system assigned the following weights:

- Screen Time: 40%
- Data Usage: 30%
- Number of Applications Used: 30%

Based on this score, users were classified into three categories:

- Low Risk
- Medium Risk
- High Risk

This classification helped simplify the interpretation of user behavior and allowed the model to better identify patterns of intensive device usage.

Demographic segmentation was also performed by analyzing behavioral patterns across different age groups (Gen Z, Millennials, and Gen X) as well as gender categories.

3. Machine Learning Modeling

After feature engineering, the dataset was prepared for machine learning analysis. Since the dataset contained categorical variables such as device model and operating system, these variables were transformed into numerical form using **One-Hot Encoding** (`pd.get_dummies`).

A **Random Forest Classifier** was selected as the predictive model for this project. Random Forest is an ensemble learning algorithm that builds multiple decision trees and combines their predictions to improve accuracy and reduce overfitting. This method is particularly effective for handling complex relationships and nonlinear patterns within behavioral data.

The model was trained to predict user behavior risk categories based on the available features.

To evaluate the performance of the model, several metrics were used, including:

- Accuracy
- Precision
- Recall
- F1-Score

A confusion matrix and classification report were generated to assess how well the model correctly classified each category of users. The model achieved an accuracy of **99.2%**, indicating strong predictive performance for identifying user behavior patterns.

4. Data Export and Business Intelligence Integration

After completing the analysis and modeling stages, the final dataset was exported as **Final_User_Behavior_Analysis.csv**.

This dataset includes both the original variables and the newly engineered features such as burnout scores and risk categories. The exported file was then integrated into **Power BI** to create interactive dashboards for visualization and reporting.

The Power BI dashboard allows users to explore key metrics related to device usage, battery consumption, and behavioral patterns through dynamic visualizations.

Key Analytical Insights

The analysis produced several important insights about user behavior and mobile device usage.

First, there is a **clear positive relationship between application usage time and battery consumption**. As screen time increases, battery drain increases at a relatively predictable rate.

Second, users naturally fall into different **usage tiers**. Some users demonstrate low engagement with minimal battery consumption, while others show significantly higher usage intensity and battery drain.

The trend line observed in the scatter plot represents the **average battery consumption per minute of usage**. Users located above the trend line tend to be more efficient, achieving more usage time for the battery consumed. In contrast, users below the line experience higher battery drain for the same amount of usage time.

Another important finding is the presence of a **behavioral threshold at approximately 300 minutes of daily usage**. Beyond this point, Digital Burnout scores increase significantly, suggesting that around five hours of daily screen time may represent a critical threshold for intensive device usage.

Demographic analysis also revealed that although **Gen X users represent a larger share of the dataset**, **Gen Z users demonstrate higher average engagement levels**, with the highest app usage time per user.

Finally, the results suggest that **battery drain is influenced more by user behavior than by device hardware**. Across different device models, battery consumption patterns remain relatively consistent, while usage intensity is the main factor driving energy consumption.